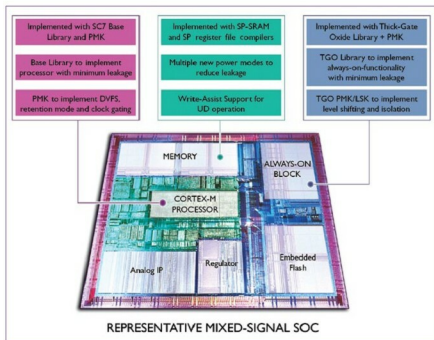


Outlook 2014: Embedding Digital Processing in Analogue and Mixed-Signal Designs

In the coming years, as the IoT becomes a larger reality and the demand for reduced power consumption heightens, the model for the next-generation sophisticated analogue and mixed-signal circuits is likely to combine 'analogue on the edge'—to sense and actuate—with 'digital at the heart'—to control, make decisions and communicate



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The market for analogue and mixed-signal semiconductors is set for some significant evolution over the next few years. The Internet of Things—in which billions of objects (smart or otherwise) will become IP enabled and connected—offers an abundance of opportunities for devices and systems that sense, control, actuate and communicate with the physical and most decidedly analogue world in which we live.

Another important trend that is uppermost in the collective mind of the entire electronics industry is delivering solutions that provide reduced power consumption and increased energy efficiency at the system level. Could the encroachment of digital-processing technology into the esoteric field of analogue chip design provide an answer for each of these strategic trends?

Analogue design is, of course, the proverbial black art and combining it with digital is a highly complex pro-

cess in advanced mixed-signal designs. Arguably, though, it has become much easier in recent times. One important factor that has eased and sped up the development process is the growing sophistication of the mixed-signal design flow with EDA tools that now allow engineers to design and simulate both the analogue and digital in mixed-signal designs, whilst running C code simultaneously.

Previously, there have always been varying degrees of separation between the design and simulation of analogue and digital circuitry; and when it comes together in a single-chip layout, the design might not always work first time. But increasingly, thanks to these new advanced design tools, the process is now almost instantaneous, rather than taking many extra weeks in development. And when the increasing commercial availability of analogue IP along with digital libraries is added into the mix, this makes it significantly easier and quicker for any analogue or mixed-signal chip vendor to come up with fast and efficient designs in a way that was not possible a few years ago—and certainly for those without huge experience in this field.

Embedding processing capability

These advanced capabilities in the design flow make it increasingly easy to embed a digital processor into an ana-